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RESEARCH NOTE

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Opportunistic foraging strategy of rainbow lizards at a seaside resort in Togo

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Abstract

Rainbow lizards (*Agama agama*) are common in suburban areas throughout Africa, and have an opportunistic foraging strategy, with arthropods being the main prey source. In a coastal resort in southern Togo, West Africa, several individuals in a population were observed while feeding regularly upon non-natural human-made food (pizza) and showing a clear preference for a given type of food versus others that were offered ('four cheeses' being the preferred one). The fact that all monitored individuals fed upon a same type of pizza suggests that they may have some chemical cues attracting them.

KEYWORDS
Agamidae, diet, suburban, West Africa

Résumé

Les lézards arc-en-ciel (*Agama agama*) sont présents partout en Afrique dans les zones suburbaines et ont une stratégie de recherche de nourriture opportuniste, les arthropodes étant leur principale source de proies. Dans une station balnéaire du sud du Togo, en Afrique de l'Ouest, on a observé que plusieurs individus d'une population se nourrissaient régulièrement d'aliments non naturels fabriqués par l'homme (pizzas) et montraient une nette préférence pour un type d'aliment donné par rapport aux autres qui leur étaient proposés (la pizza « quatre fromages » était la préférée). Comme tous les individus observés se nourrissaient du même type de pizza, on peut penser qu'ils sont attirés par des signaux chimiques.

Rainbow lizards (*Agama agama*) are among the most common lizard species in urban and suburban areas in West Africa, including Togo (Luiselli et al., 2022). They are also well known for their ecological plasticity in both thermal ecology (Amadi et al., 2020) and foraging strategy. This includes the ability of some individuals to shift from operating as strictly diurnal in rural areas, to nocturnal foragers of insects and other arthropods nearby using artificial lights as a heat source in effect, a gecko-like foraging behaviour (Amadi et al., 2021). Rainbow lizards are predominantly insectivorous (Akani et al., 2013; Anibaldi et al., 1998), but some populations may also show an omnivorous diet (Pauwels et al., 2017). These lizards have been reported to feed also upon bread-based food in the surroundings of human settlements (Pauwels & David, 2008; Yeboah Ofori et al., 2018). Here, we describe a previously unreported bread-based food type and an opportunistic foraging strategy by these normally insectivorous lizards in a seaside resort in Togo, West Africa.

The observations were conducted at the 'Pure Plage', a touristic seaside resort in Baguida, Lomé, during October 2020 and April–May 2021. All observations were made between 12:00 h and 15:00 h (Lomé time), during sunny days with ambient temperatures of 31–33°C. During this time, we monitored the behaviour of nine *A. agama* individuals, including five adults (three males and two females) and four subadults/juveniles. After having opportunistically observed an adult male 'stealing' a piece of four cheese pizza from tourists by climbing a table, we decided to investigate whether this behaviour was occasional or usual in this rainbow lizard population. To do this

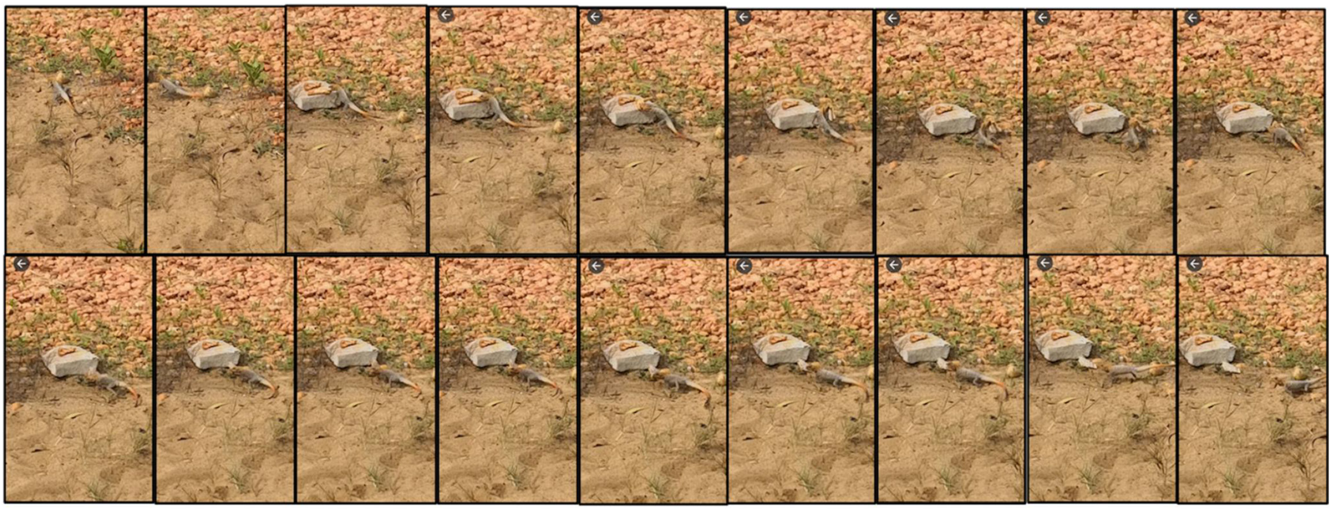

FIGURE 1 Rainbow lizards eating 'four cheese' pizza at a seaside touristic resort in Togo.

during two different days, we offered at the same time two different plates of pizza (one 'four cheeses' and one 'four seasons') placed on the ground, spaced about 10 m apart each from the other and about 10 m from trees where the adult lizards were previously spotted, and monitored their behaviour from a distance of about 15 m from the spots where pizzas were placed.

Within 15 min from the start of the experiment, lizards actively fed upon pizza pieces (Figure 1), including fighting over the food, and even running to a feeding location from as far as 15 m, and on multiple occasions. The behaviour lasted for about 120 min, with all lizards eating the pizza pieces but only the 'four cheeses' type.

Although essentially insectivorous (Akani et al., 2013), rainbow lizards have been occasionally observed to feed upon bread pieces (Pauwels & David, 2008; Yeboah Ofori et al., 2018). However, this is the first study documenting several individuals in a population feeding regularly upon non-natural human-made food and showing, a preference for a given type of food versus others that were offered. The fact that all monitored individuals fed upon a same type of pizza in preference the other type suggests that they may have some chemical cues attracting them, or that the specific 'four cheeses' type may provide nutrients that more easily digested. Future studies should investigate the possibility that this lizard population consuming an alternative (non-natural) type of food may be in better body condition than other individuals, from a same geographic area, exhibiting a typical insectivorous diet.

CONFLICT OF INTEREST

None.

DATA AVAILABILITY STATEMENT

All data that support the findings of this study are presented in the manuscript.

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